

Transition Metals

General properties of transition metals	know that transition metal characteristics of elements Ti – Cu arise from an incomplete d sub-level in atoms or ions
	know that these characteristics include complex formation, formation of coloured ions, variable oxidation state and catalytic activity
Complex formation	be able to define the term <i>ligand</i>
	know that co-ordinate bonding is involved in complex formation
	understand that a complex is a central metal ion surrounded by ligands
	know the meaning of co-ordination number
	understand that ligands can be unidentate (e.g. H ₂ O, NH ₃ and Cl ⁻)
	or bidentate (e.g. NH ₂ CH ₂ CH ₂ NH ₂ and C ₂ O ₄ ²⁻)
	or multidentate (e.g. EDTA ⁴⁻)
Shapes of complex ions	know that haem is an iron(II) complex with a multidentate ligand
	know that transition metal ions commonly form octahedral complexes with small ligands (e.g. H ₂ O and NH ₃)
	know that transition metal ions commonly form tetrahedral complexes with larger ligands (e.g. Cl ⁻)
	know that square planar complexes are also formed, e.g. cisplatin
Formation of coloured ions	know that Ag ⁺ commonly forms the linear complex [Ag(NH ₃) ₂] ⁺ as used in Tollens' reagent
	know that transition metal ions can be identified by their colour, limited to the complexes in this unit
	know that colour changes arise from changes in oxidation state, co-ordination number and ligand
	know that colour arises from electronic transitions from the ground state to excited states: $\Delta E = h\nu$
Variable oxidation states	appreciate that this absorption of visible light is used in spectrometry to determine the concentration of coloured ions
	know that transition elements show variable oxidation states
	know that Cr ³⁺ and Cr ²⁺ are formed by reduction of Cr ₂ O ₇ ²⁻ by zinc in acid solution
	know the redox titration of Fe ²⁺ with MnO ₄ ⁻ and Cr ₂ O ₇ ²⁻ in acid solution
	be able to perform calculations for this titration and for others when the reductant and its oxidation product are given
	know the oxidation of Co ²⁺ by air in ammoniacal solution
	know the oxidation in alkaline solution of Co ²⁺ and Cr ³⁺ by H ₂ O ₂

Catalysis	know that transition metals and their compounds can act as heterogeneous and homogeneous catalysts
Heterogeneous	know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction occurs at the surface
	understand the use of a support medium to maximise the surface area and minimise the cost (e.g. Rh on a ceramic support in catalytic converters)
	understand how V_2O_5 acts as a catalyst in the Contact Process
	know that a Cr_2O_3 catalyst is used in the manufacture of methanol from carbon monoxide and hydrogen
	know that Fe is used as a catalyst in the Haber Process
	know that catalysts can become poisoned by impurities and consequently have reduced efficiency; know that this has a cost implication (e.g. poisoning by sulfur in the Haber Process and by lead in catalytic converters in cars)
Homogeneous	know that when catalysts and reactants are in the same phase, the reaction proceeds through an intermediate species (e.g. the reaction between I^- and $S_2O_8^{2-}$ catalysed by Fe^{2+} and autocatalysis by Mn^{2+} in reactions of $C_2O_4^{2-}$ with MnO_4^-).
Other applications of transition metal complexes	understand the importance of variable oxidation states in catalysis; both heterogeneous and homogeneous catalysts
	understand that Fe(II) in haemoglobin enables oxygen to be transported in the blood, and why CO is toxic
	know that the Pt(II) complex <i>cisplatin</i> is used as an anticancer drug
	appreciate the benefits and risks associated with this drug
	understand that $[Ag(NH_3)_2]^+$ is used in Tollens' reagent to distinguish between aldehydes and ketones